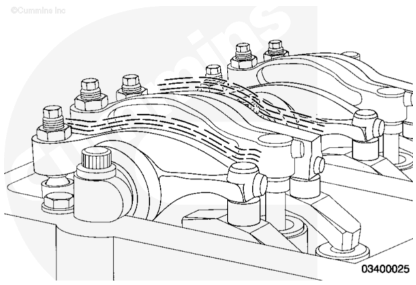
Trainings ▾

General Information

with Mechanically Actuated Injector

The valves and injectors **must** be correctly adjusted for the engine to operate efficiently. Valve and injector adjustment **must** be performed using the values listed in this procedure.

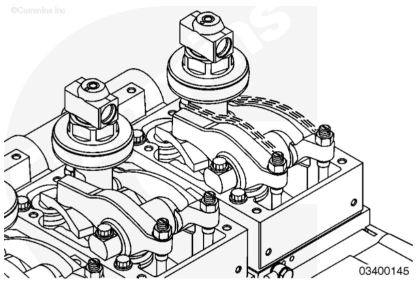


Cummins Inc. has found that engines in most applications will **not** experience significant valve/injector train wear. It is recommended valve and injector adjustment is performed **only** when the injectors are removed or when other repairs disturb the valve train.

If valve and injector adjustment is checked during troubleshooting or before the recommended maintenance interval, adjustment is **not** required if measurements are within the recheck limits.

with Electronically Actuated Injector

The valves **must** be correctly adjusted for the engine to operate efficiently. Valve adjustment **must** be performed using the values listed in this procedure.



For engines with electronically actuated injectors, periodic valve adjustment is **not** required. It is recommended that the valves be adjusted **only** when an injector is removed.

Preparatory Steps

with Mechanically Actuated Injector

- Remove the rocker lever cover and all related components. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-om.html) with Electronically Actuated Injector

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

⚠ WARNING ⚠

Pressure within the high-pressure fuel system must never be measured using a mechanical gauge. Fuel pressure values of over 1724 bar [25,000 psi] are possible. Under high-pressure, a mechanical gauge can fail causing a high-pressure fuel leak resulting in personal injury and property damage.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

⚠ CAUTION ⚠

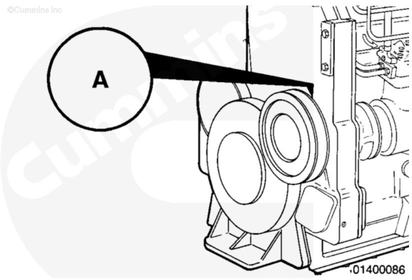
A very small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation.

- Clean all fittings and components with steam or solvent before removal from the engine.
- Connect INSITE™ electronic service tool and verify that the fuel pressure has bled down by monitoring the fuel pressure.
- Remove the injector supply line and supply lines. Refer to Procedure 006-051 in Section A. (/qs3/pubsys2/xml/en/procedures/20/20-006-051-om.html)
- Remove rocker lever cover. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-om.html)

Adjust

with Mechanically Actuated Injector

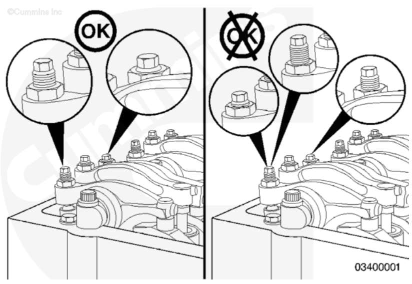
Note : The barring device shaft turns approximately two revolutions before the engine begins to turn. The barring device will not turn the engine opposite the direction of normal rotation.



Push the shaft in and turn the barring device **counterclockwise** until the "A" mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.

Determine the Cylinder in Position for Valve Set.

Valve Set Mark	Check Valve Position
A	1, 6
B	2, 5
C	3, 4



If the rocker lever assemblies have been removed, use this step to determine the cylinder to be set.

Lubricate the adjusting screw threads with clean engine oil prior to making valve and injector adjustments.

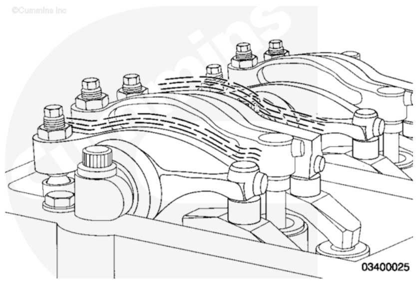
All adjusting screws **must** be loose on all cylinders, and the push rods **must** remain in alignment.

Perform this step on both cylinders to be checked.

Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.

The push rods will be the same height above the top of the rocker lever housing on the cylinder ready for valve adjustment.

If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question. Use the chart and select the correct cylinder for adjustment.

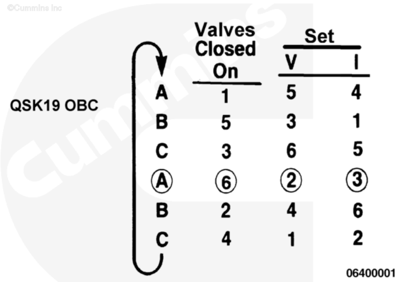


Use the chart to determine the valve and injector that is ready to adjust.

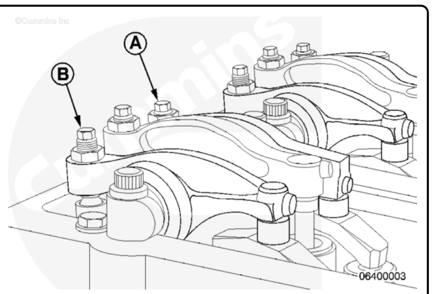
Adjustment can begin on any valve set mark.

In the example, assume the A mark is aligned and the push rod heights indicate that one valve on cylinder number 1 is fully opened and both valves on cylinder number 6 are fully closed. The chart also shows that the valves on cylinder number 2 and the injector on cylinder 3 are ready to adjust.

After the adjustment, bar the engine to the B set mark. Adjust the valves on cylinder number 4 and adjust the injector on cylinder number 6.

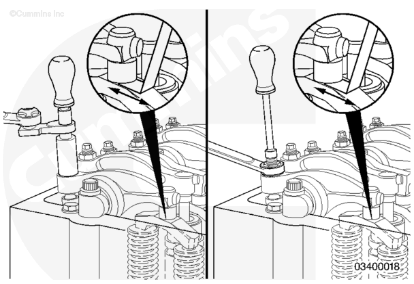


Valve Adjustment (Initial Set)			
Reference Point	**mm**		**in**
A	0.81	Exhaust	0.032
B	0.36	Intake	0.014



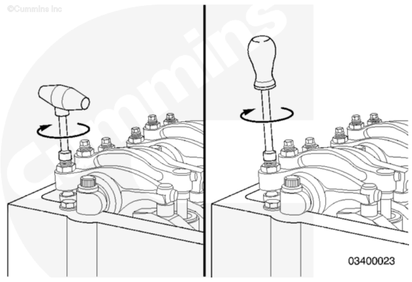
Make certain the crosshead is firmly in place on the valve stem tips.

Make certain the feeler gauge is under the center of the ball and socket, or the socket can rock or tip, resulting in an incorrect adjustment. To avoid false readings, hold the swivel foot flat to avoid binding while checking the lash.



Use service tool Part Number 3824901. Select a feeler gauge for the correct valve lash specification. Insert the gauge between the rocker lever socket and the crosshead.

Two different methods for establishing valve lash clearance are described below. Either method can be used. The torque wrench method has proven to be the **most** consistent.



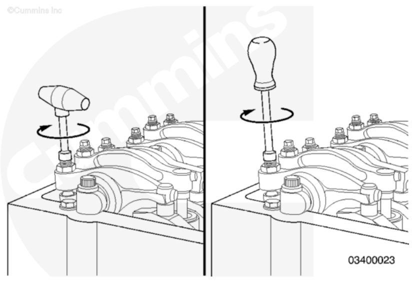
Torque Wrench Method:

- Use service tool, Part Number 3376592, inch-pound torque wrench, to tighten the adjusting screw to .68 N•m [6 in-lb] torque against the feeler gauge.

Feel Method:

- Use a nut driver and turn the adjusting screw **only** until the lever touches the feeler gauge.

Be certain the parts are in alignment. Tighten the adjusting screw and squeeze the oil out of the valve train.



Loosen the adjusting screw at least one full revolution.

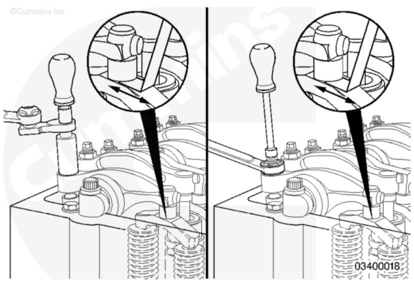
Insert a feeler gauge between the rocker lever socket and the crosshead.

Use torque wrench, Part Number 3376592, and tighten the adjustment screw.

Torque Value: 0.68 n•m [6 in-lb]

Remove feeler gauge.

The adjustment screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.



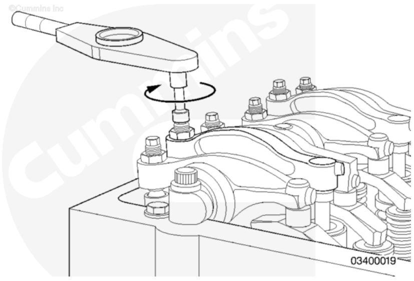
Tighten the locknut.

Torque Value	
Adjustment Screw Part Number	168306
With Adapter	48 N•m [35 ft-lb]
Without Adapter	60 N•m [44 ft-lb]
Adjustment Screw Part Number	3090007
With Adapter	84 N•m [62 ft-lb]
Without Adapter	105 N•m [77 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the thicker gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

Use a torque wrench to tighten the injector rocker lever adjusting screw. If the screw chatters during setting, repair the screw and lever as required.



Be certain the parts are in alignment. Tighten the adjusting screw and squeeze the oil out of the injector train. This is an initial adjustment to preload the injector.

Torque Value: 28 n•m [248 in-lb]

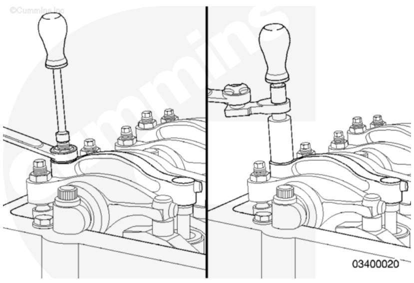
Loosen the adjusting screw at least one revolution.

Tighten the adjusting screw again to the final setting.

Torque Value: 19 n•m [168 in-lb]

The torque wrench **must** be calibrated, have a resolution of 0.28 N•m [2.5 in-lb], and have a range of 17 to 23 N•m [150 to 200 in-lb].

Hold the adjusting screw in this position. The adjusting screw **must not** turn when the locknut is tightened.



Tighten the locknut to the following values:

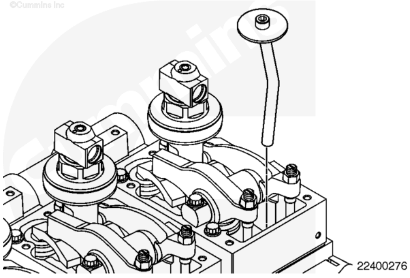
For the torque method (with adapter), use torque wrench adapter, Part Number ST-669.

Torque Value	
Adjustment Screw Part Number	168306
With Adapter	48 N•m [35 ft-lb]
Without Adapter	60 N•m [44 ft-lb]
Adjustment Screw Part Number	3090007
With Adapter	84 N•m [62 ft-lb]
Without Adapter	105 N•m [77 ft-lb]

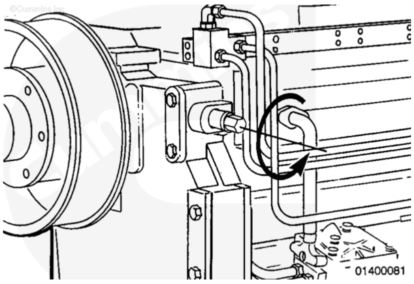
with Electronically Actuated Injector

⚠ CAUTION ⚠

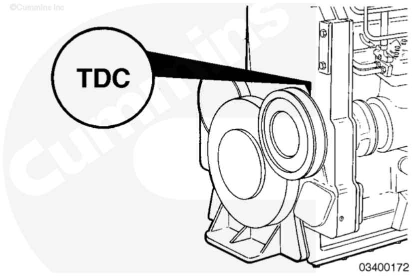
To reduce the possibility of engine damage, use a cylinder head protective cover to prevent tools from falling into the cam follower cavity.



Install the cylinder head protective cover, Part Number 4918282, into the push tube hole.

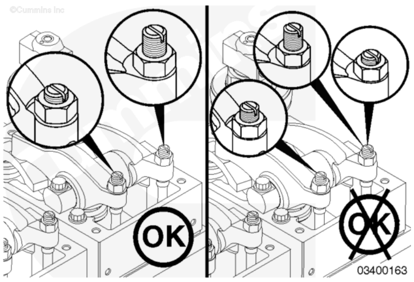


The barring device shaft turns approximately two revolutions before the engine begins to turn. The barring device will **not** turn the engine opposite the direction of normal rotation.



TDC represents Top Dead Center for cylinders 1 or 6. This mark will be used to set the valves on engines with electronically actuated injectors. This will allow all valves to be set in two positions. Ignore the A, B, and C marks while using this procedure.

Push the shaft in and turn the barring device until the TDC mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.



The number of threads visible above the adjusting nut will **not** be the same. There will be more threads visible on the intake adjusting screw than on the exhaust adjusting screw.

If the rocker lever assemblies have been removed, use this step to determine the cylinder to be set.

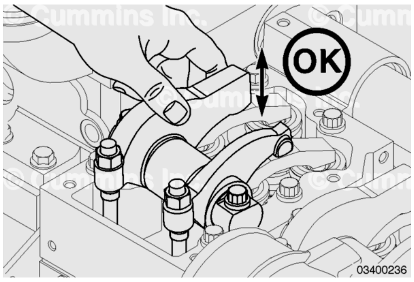
All adjusting screws **must** be loose on all cylinders, and the push rods **must** remain in alignment.

Perform this step on both cylinders to be checked.

Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.

The push rods will be the same height above the top of the rocker lever housing on the valves that should be loose.

If both number 1 cylinder rocker levers are loose, proceed to the next step. If the number 1 cylinder rocker levers are **not** loose, rotate the crankshaft 360 degrees and proceed to the next step.

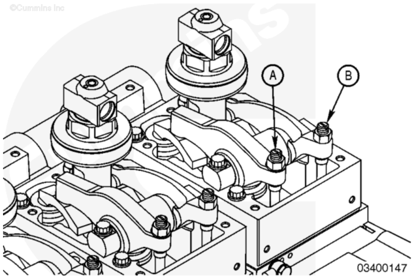


If the number 1 cylinder is at TDC and both rocker levers are loose, the valve lash (overhead set) can be checked on the following rocker levers:

- E = exhaust
- I = intake

1I, 1E, 2I, 3E, 4I, and 5E

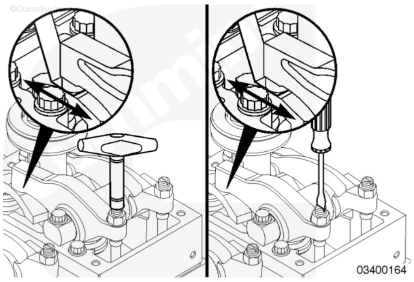
Valve Adjustment (Initial Set)	
Exhaust (A)	**Intake (B)**
0.69 mm [0.027 in]	0.36 mm [0.014 in]



Valve Recheck Limits

	mm		in
Intake Valve	0.280	MIN	0.011
	0.430	MAX	0.017
Exhaust Valve	0.610	MIN	0.027
	0.762	MAX	0.030

Use service tool, Part Number 3163171 (intake) or Part Number 3163172 (exhaust). Select a feeler gauge for the correct valve lash specification. Insert the gauge between the rocker lever socket and the crosshead.



Make certain the crosshead is firmly in place on the valve stem tips.

Make certain the feeler gauge is under the center of the ball and socket, or the socket can rock or tip, resulting in an incorrect adjustment. To avoid false readings, hold the swivel foot flat to avoid binding while checking the lash.

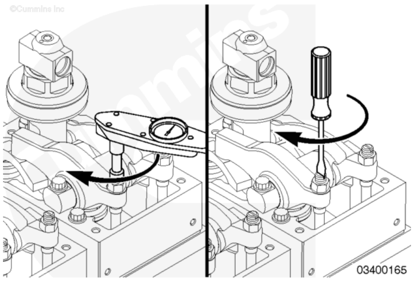
Two different methods for establishing valve lash clearance are described below:

- Torque Wrench Method - Use Part Number 3376592, inch-pound torque wrench, to tighten the adjusting screw to

0.68 N•m [6 in-lb] torque against the feeler gauge.

- Feel Method - Use a screwdriver and turn the adjusting screw **only** until the lever touches the feeler gauge.

Either method can be used. The torque wrench method has proven to be the most consistent.



To set the valves using the torque wrench method, complete the following steps:

Be certain the parts are in alignment. Tighten the adjusting screw and squeeze the oil out of the valve train.

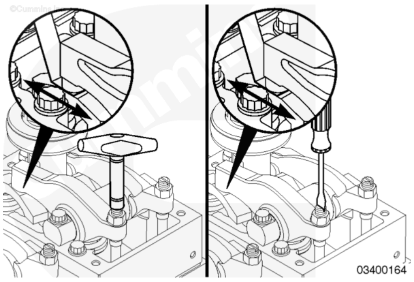
Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever socket and the crosshead.

Use torque wrench, Part Number 3376592, and tighten the adjusting screw.

Remove the feeler gauge.

Torque Value: 0.68 n•m [6 in-lb]



To set the valves using the feel method, complete the following steps:

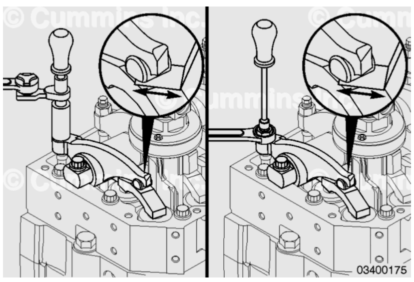
The adjustment screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.

Tighten the locknut.

The adjusting screw **must not** turn when the locknut is tightened.

For the torque method (with adapter), use torque wrench adapter, Part Number ST-669.

Torque Value	
Adjustment Screw Part Number	168306
With Adapter	48 N•m [35 ft-lb]
Without Adapter	60 N•m [44 ft-lb]
Adjustment Screw Part Number	3090007
With Adapter	84 N•m [62 ft-lb]



Torque Value	
Without Adapter	105 N•m [77 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the thicker gauge will fit.

Repeat the adjustment process until the proper clearance is obtained.

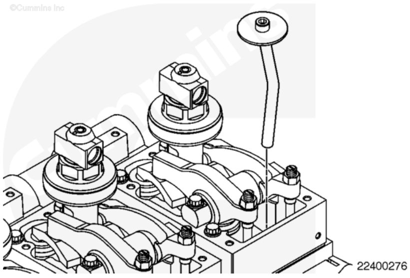
Use the barring tool to rotate the crankshaft 360 degrees. Use the previous steps and specifications to set the valve lash on the following rocker levers:

- E = exhaust
- I = intake

2E, 3I, 4E, 5I, 6I, 6E.

If the measurements are out of specification, set the valve lash.

Remove the cylinder head protective cover, Part Number 4918282, from the push tube hole.



Finishing Steps

with Mechanically Actuated Injector

- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-om.html)
- Operate the engine and check for leaks.

with Electronically Actuated Injector

⚠ **WARNING** ⚠

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⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

⚠ WARNING ⚠

Pressure within the high-pressure fuel system must never be measured using a mechanical gauge. Fuel pressure values of over 1724 bar [25,000 psi] are possible. If a mechanical gauge is used it can fail, causing a high-pressure fuel leak which can cause personal injury and property damage.

⚠ CAUTION ⚠

A very small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation.

- Install rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-tr.html)
- Install rocker lever cover. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/20/20-003-011-om.html)
- Install new injector supply line and supply lines between injectors. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-051-tr.html)
- Install new injector supply line and supply lines between injectors. Refer to Procedure 006-051 in Section A. (/qs3/pubsys2/xml/en/procedures/20/20-006-051-om.html)
- Operate the engine and check for leaks.